Bài 2. Trao đổi dữ liệu sử dụng giao thức MQTT

Yêu cầu. Viết chương trình (bằng ngôn ngữ tùy ý: C#, **Java**, python) thực hiện

- Gửi (publish) dữ liệu lên một MQTT broker. Ví dụ dùng broker: **broker.hivemq.com:1883**
- Đóng gói dữ liệu bằng JSON. Ví dụ:

```
{
    "DeviceName": "my-mqtt-client",
    "temperature":30,
    "humidity":60
}
```

- Nhận (subscribe) dữ liệu từ broker.
- Parse dữ liệu nhận được và hiển thị các trường dữ liệu tương ứng.
- Sử dụng công cụ MQTTBox để kiểm tra kết quả chạy chương trình.

Yêu cầu nộp bài:

Viết báo cáo mô tả ngắn gọn gồm: code chính của chương trình, ***chụp ảnh màn hình thực hiện chương trình và kết quả ở các yêu cầu***.

Submit: file báo cáo, file code

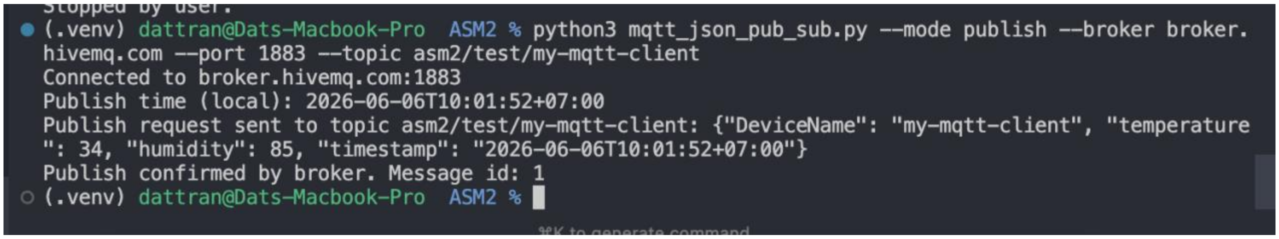
Code chính:

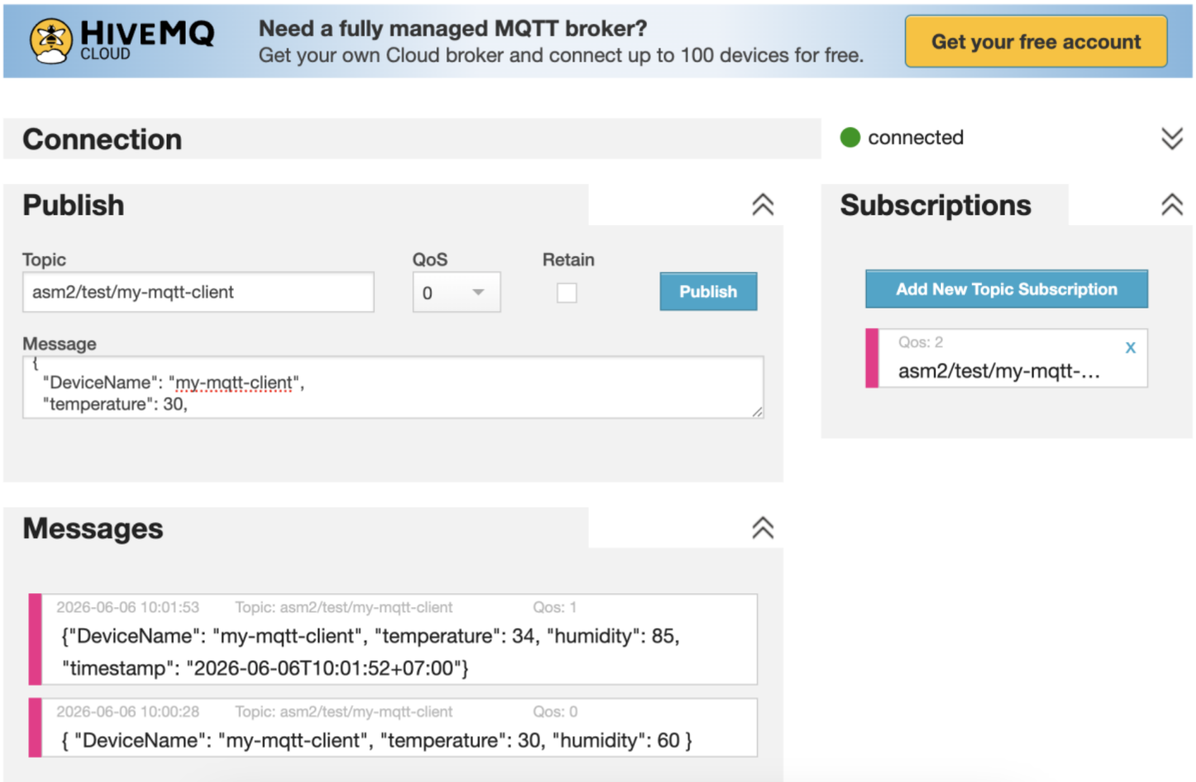
```python
def publish_payload(client, args, payload, message_received):
    message_payload = payload.copy()
    message_payload["timestamp"] = local_timestamp()
    json_payload = json.dumps(message_payload)
    result = client.publish(args.topic, json_payload, qos=1)
    if result.rc != mqtt.MQTT_ERR_SUCCESS:
        print(f"Publish request failed with result code: {result.rc}")
        message_received.set()
        return

    print(f"Publish time (local): {message_payload['timestamp']}")
```

```
    print(f"Publish request sent to topic {args.topic}: {json_payload}")


def build_client(args, payload, message_received):
    client_id = f"python-mqtt-json-{uuid.uuid4().hex[:8]}"
    client = mqtt.Client(
        callback_api_version=mqtt.CallbackAPIVersion.VERSION2,
        client_id=client_id,
    )

    def on_connect(client, userdata, flags, reason_code, properties):
        if reason_code != 0:
            print(f"Connection failed with reason code: {reason_code}")
            message_received.set()
            return

        print(f"Connected to {args.broker}:{args.port}")
        if args.mode == "publish":
            publish_payload(client, args, payload, message_received)
            return

        result, message_id = client.subscribe(args.topic)
        if result != mqtt.MQTT_ERR_SUCCESS:
            print(f"Subscribe request failed with result code: {result}")
            message_received.set()
            return

        print(f"Subscribe request sent for topic: {args.topic}")

    def on_subscribe(client, userdata, message_id, reason_codes, properties):
        print(f"Subscribed to topic: {args.topic}")

        if args.mode == "both":
            publish_payload(client, args, payload, message_received)

    def on_publish(client, userdata, message_id, reason_code, properties):
        print(f"Publish confirmed by broker. Message id: {message_id}")
```

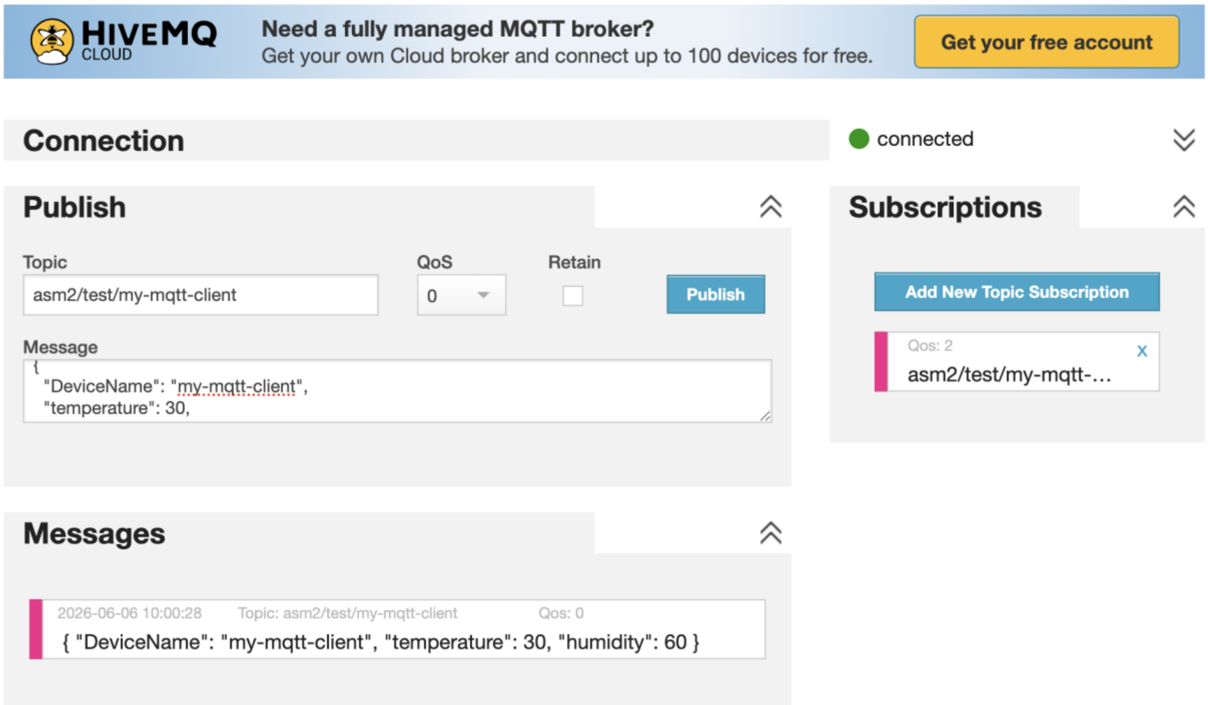
```
    if args.mode == "publish":
        message_received.set()

def on_message(client, userdata, message):
    print(f"\nMessage received from topic: {message.topic}")
    print(f"Receive time (local): {local_timestamp()}")
    try:
        decoded_message = message.payload.decode("utf-8")
        received_payload = json.loads(decoded_message)
    except UnicodeDecodeError:
        print("Could not decode MQTT payload as UTF-8 text.")
    except json.JSONDecodeError:
        print("Could not parse MQTT payload as JSON.")
        print(f"Raw payload: {message.payload!r}")
    else:
        display_fields(received_payload)
    finally:
        if args.mode != "subscribe" or args.timeout > 0:
            message_received.set()

def on_disconnect(client, userdata, disconnect_flags, reason_code, properties):
    if reason_code != 0 and not message_received.is_set():
        print(f"Disconnected before receiving data. Reason code: {reason_code}")
        message_received.set()

client.on_connect = on_connect
client.on_subscribe = on_subscribe
client.on_publish = on_publish
client.on_message = on_message
client.on_disconnect = on_disconnect
return client
```

Gửi (publish) dữ liệu tới broker





Nhận (subcribe) dữ liệu từ broker




```
Stopped by user.
(.venv) dattran@Dats-Macbook-Pro ASM2 % python3 mqtt_json_pub_sub.py --mode subscribe --broker broke
r.hivemq.com --port 1883 --topic asm2/test/my-mqtt-client --timeout 0
Waiting for MQTT messages. Press Ctrl+C to stop.
Connected to broker.hivemq.com:1883
Subscribe request sent for topic: asm2/test/my-mqtt-client
Subscribed to topic: asm2/test/my-mqtt-client

Message received from topic: asm2/test/my-mqtt-client
Receive time (local): 2026-06-06T10:00:28+07:00
Retrieved data:
DeviceName  : my-mqtt-client
temperature : 30
humidity    : 60
```